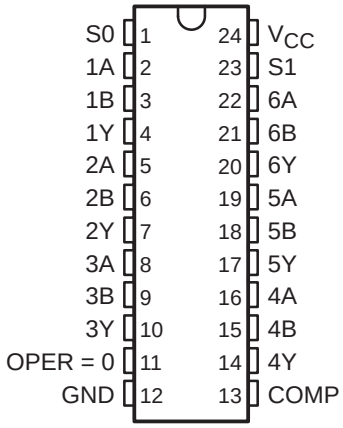
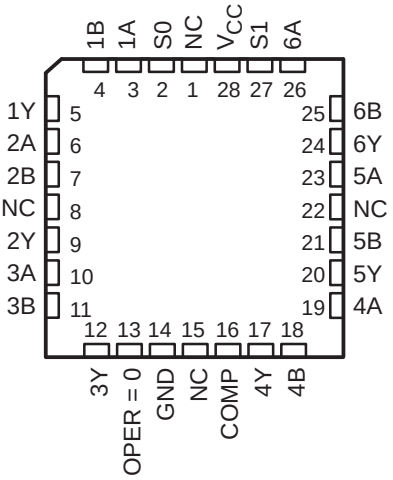
- **Select True or Complementary Data**
- **Perform AND/NAND (Masking) of A or B Operand**
- **Cascadable to Expand Number of Operands**
- **Detect Zeros on A or B Operands**
- **3-State Outputs Interface Directly With System Bus**
- **Package Options Include Plastic Small-Outline (DW) Packages, Ceramic Chip Carriers (FK), and Standard Plastic (NT) and Ceramic (JT) 300-mil DIPs**

SN54ALS857 . . . JT PACKAGE
SN74ALS857 . . . DW OR NT PACKAGE
(TOP VIEW)



SN54ALS857 . . . FK PACKAGE
(TOP VIEW)



NC – No internal connection

description

The ′ALS857 are hextuple 2-line to 1-line multiplexers with 3-state outputs. The devices can provide either true (COMP low) or inverted (COMP high) data at the Y outputs. In addition, the ′ALS857 perform the logical AND function (A • B) and the clear function as well. The four modes of operation are:

- Select A-data inputs
- Select B-data inputs
- AND A inputs with B inputs
- Clear

In either of the first two modes, OPER = 0 is high if all the selected A or B inputs are low. The six Y outputs and the OPER = 0 output are all 3-state and rated at 12-mA and 24-mA I_{OL} for the SN54ALS857 and SN74ALS857, respectively. All outputs can be placed in the high-impedance state by applying a high level to the COMP, S0, and S1 inputs simultaneously.

The SN54ALS857 is characterized for operation over the full military temperature range of –55°C to 125°C. The SN74ALS857 is characterized for operation from 0°C to 70°C.

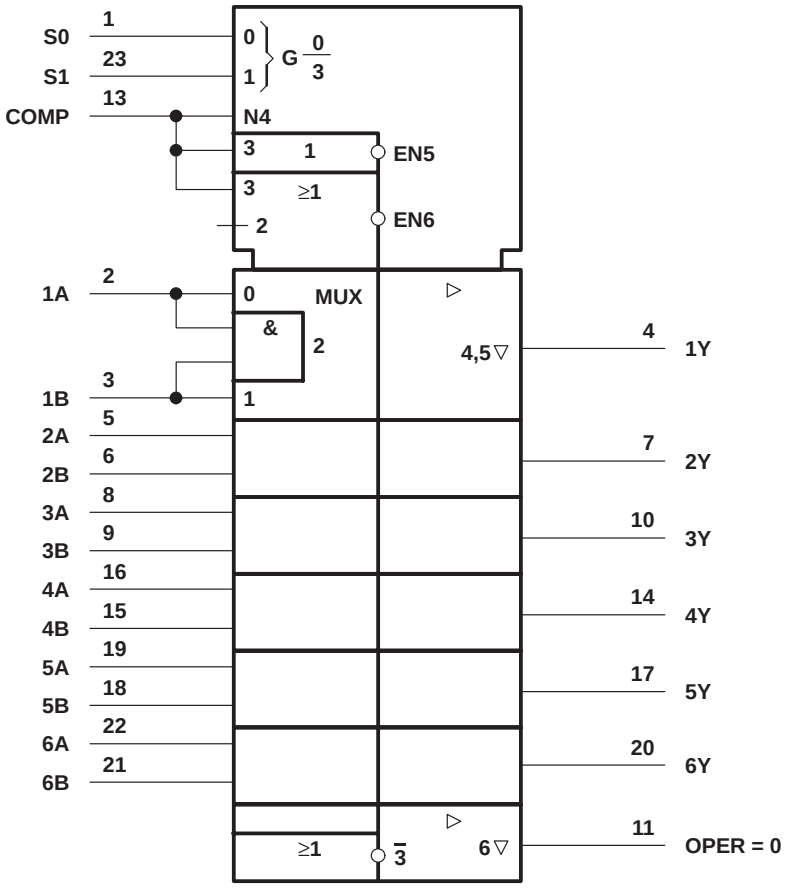
PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

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FUNCTION TABLE

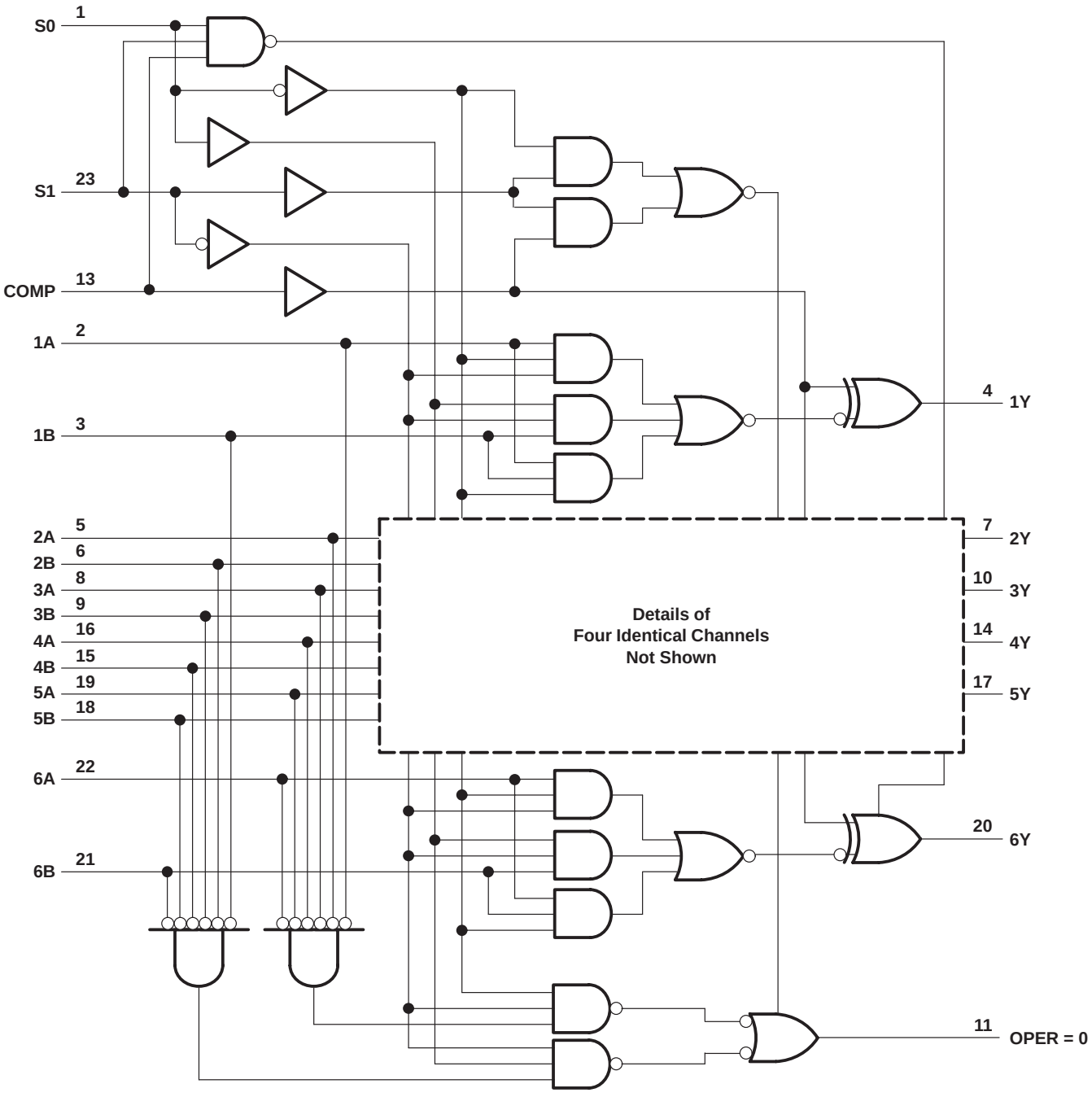
INPUTS			OUTPUTS	
COMP	S1	S0	Y	OPER = 0
L	L	L	A	H = all A inputs L
L	L	H	B	H = all B inputs L
L	H	L	A • B	Z
L	H	H	L	L
H	L	L	$\overline{A}$	H = all A inputs L
H	L	H	$\overline{B}$	H = all B inputs L
H	H	L	$\overline{A \bullet B}$	Z
H	H	H	Z	Z

logic symbol†



† This symbol is in accordance with ANSI/IEEE Std 91-1984 and IEC Publication 617-12.
Pin numbers shown are for the DW, JT, and NT packages.

logic diagram (positive logic)



Pin numbers shown are for the DW, JT, and NT packages.

absolute maximum ratings over operating free-air temperature range (unless otherwise noted)†

Supply voltage, V_{CC} . . . . . 7 V
Input voltage, V_I . . . . . 7 V
Voltage applied to a disabled 3-state output . . . . . 5.5 V
Operating free-air temperature range, T_A: SN54ALS857 . . . . . –55°C to 125°C
SN74ALS857 . . . . . 0°C to 70°C
Storage temperature range . . . . . –65°C to 150°C

† Stresses beyond those listed under "absolute maximum ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under "recommended operating conditions" is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

recommended operating conditions

		SN54ALS857			SN74ALS857			UNIT
		MIN	NOM	MAX	MIN	NOM	MAX	
V_{CC}	Supply voltage	4.5	5	5.5	4.5	5	5.5	V
V_{IH}	High-level input voltage	2			2			V
V_{IL}	Low-level input voltage			0.7			0.8	V
I_{OH}	High-level output current			–1			–2.6	mA
I_{OL}	Low-level output current			12			24	mA
T_A	Operating free-air temperature	–55		125	0		70	°C

electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS		SN54ALS857			SN74ALS857			UNIT
			MIN	TYP‡	MAX	MIN	TYP‡	MAX	
V_{IK}	V_{CC} = 4.5 V,	I_I = –18 mA			–1.5			–1.5	V
V_{OH}	V_{CC} = 4.5 V to 5.5 V,	I_{OH} = –0.4 mA	V_{CC} –2			V_{CC} –2			V
	V_{CC} = 4.5 V	I_{OH} = –1 mA	2.4	3.3					
		I_{OH} = –2.6 mA				2.4	3.2		
V_{OL}	V_{CC} = 4.5 V	I_{OL} = 12 mA		0.25	0.4		0.25	0.4	V
		I_{OL} = 24 mA					0.35	0.5	
I_{OZH}	V_{CC} = 5.5 V,	V_O = 2.7 V			20			20	µA
I_{OZL}	V_{CC} = 5.5 V,	V_O = 0.4 V			–20			–20	µA
I_I	V_{CC} = 5.5 V,	V_I = 7 V			0.1			0.1	mA
I_{IH}	V_{CC} = 5.5 V,	V_I = 2.7 V			20			20	µA
I_{IL}	V_{CC} = 5.5 V,	V_I = 0.4 V			–0.2			–0.2	mA
I_O§	V_{CC} = 5.5 V,	V_O = 2.25 V	–15		–70	–15		–70	mA
I_{CC}	V_{CC} = 5.5 V, See Note 1	Outputs high		11	24		11	24	mA
		Outputs low		16	33		16	33	
		Outputs disabled		18	36		18	36	

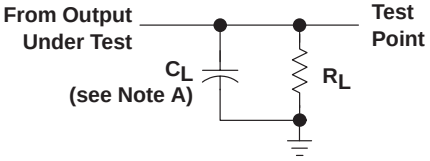
‡ All typical values are at V_{CC} = 5 V, T_A = 25°C.
§ The output conditions have been chosen to produce a current that closely approximates one half of the true short-circuit output current, I_{OS}.
NOTE 1: I_{CC} is measured with all possible inputs grounded while achieving the stated output conditions.

switching characteristics (see Figure 1)

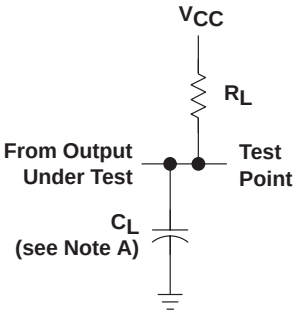
PARAMETER†	FROM (INPUT)	TO (OUTPUT)	V_{CC} = 4.5 V to 5.5 V, C_L = 50 pF, R1 = 500 Ω, R2 = 500 Ω, T_A = MIN to MAX‡				UNIT
			SN54ALS857		SN74ALS857		
			MIN	MAX	MIN	MAX	
t_{pd}	A or B (COMP high)	Y (inverting)	2	35	4	25	ns
	A or B (COMP low)	Y (noninverting)	2	27	4	18	
	S0 or S1	Y	2	37	7	33	
	COMP		2	26	6	18	
	A or B	OPER = 0	2	45	5	37	
	S0 to S1		2	30	5	23	
t_{en}	S0 to S1	Y	2	38	7	35	ns
t_{dis}			2	43	2	23	
t_{en}	COMP	Y	2	37	8	24	ns
t_{dis}			2	45	6	21	
t_{en}	S0	OPER = 0	2	29	6	20	ns
t_{dis}			2	42	11	27	
t_{en}	S1	OPER = 0	2	28	6	25	ns
t_{dis}			2	37	3	19	
t_{en}	COMP	OPER = 0	2	43	9	25	ns
t_{dis}			2	36	6	20	

† $t_{pd} = t_{PLH}$ or t_{PHL}, $t_{en} = t_{PZH}$ or t_{PZL}, $t_{dis} = t_{PHZ}$ or t_{PLZ}
‡ For conditions shown as MIN or MAX, use the appropriate value specified under recommended operating conditions.

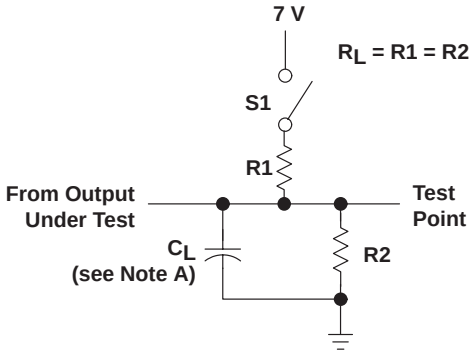
PARAMETER MEASUREMENT INFORMATION
SERIES 54ALS/74ALS AND 54AS/74AS DEVICES



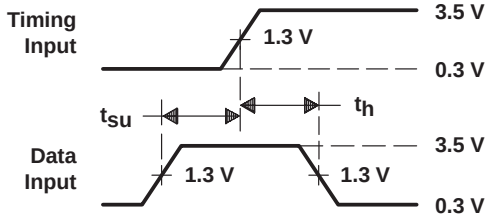
LOAD CIRCUIT FOR
BI-STATE
TOTEM-POLE OUTPUTS



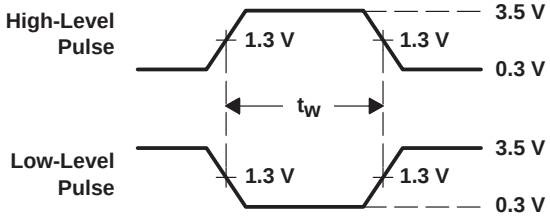
LOAD CIRCUIT
FOR OPEN-COLLECTOR OUTPUTS



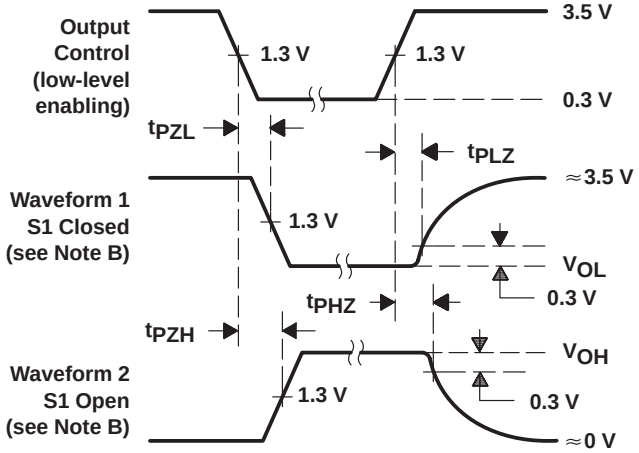
LOAD CIRCUIT
FOR 3-STATE OUTPUTS



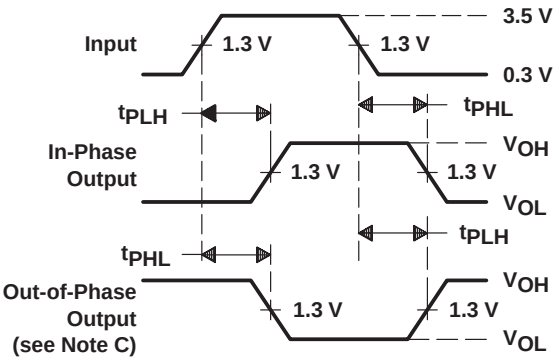
VOLTAGE WAVEFORMS
SETUP AND HOLD TIMES



VOLTAGE WAVEFORMS
PULSE DURATIONS



VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES, 3-STATE OUTPUTS



VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES

NOTES: A. C_L includes probe and jig capacitance.
B. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
C. When measuring propagation delay items of 3-state outputs, switch S1 is open.
D. All input pulses have the following characteristics: PRR ≤ 1 MHz, $t_r = t_f = 2$ ns, duty cycle = 50%.
E. The outputs are measured one at a time with one transition per measurement.

Figure 1. Load Circuits and Voltage Waveforms

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
5962-8753301LA	Active	Production	CDIP (JT) | 24	15 | TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8753301LA SNJ54ALS857JT
SNJ54ALS857JT	Active	Production	CDIP (JT) | 24	15 | TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8753301LA SNJ54ALS857JT
SNJ54ALS857JT.A	Active	Production	CDIP (JT) | 24	15 | TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8753301LA SNJ54ALS857JT

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

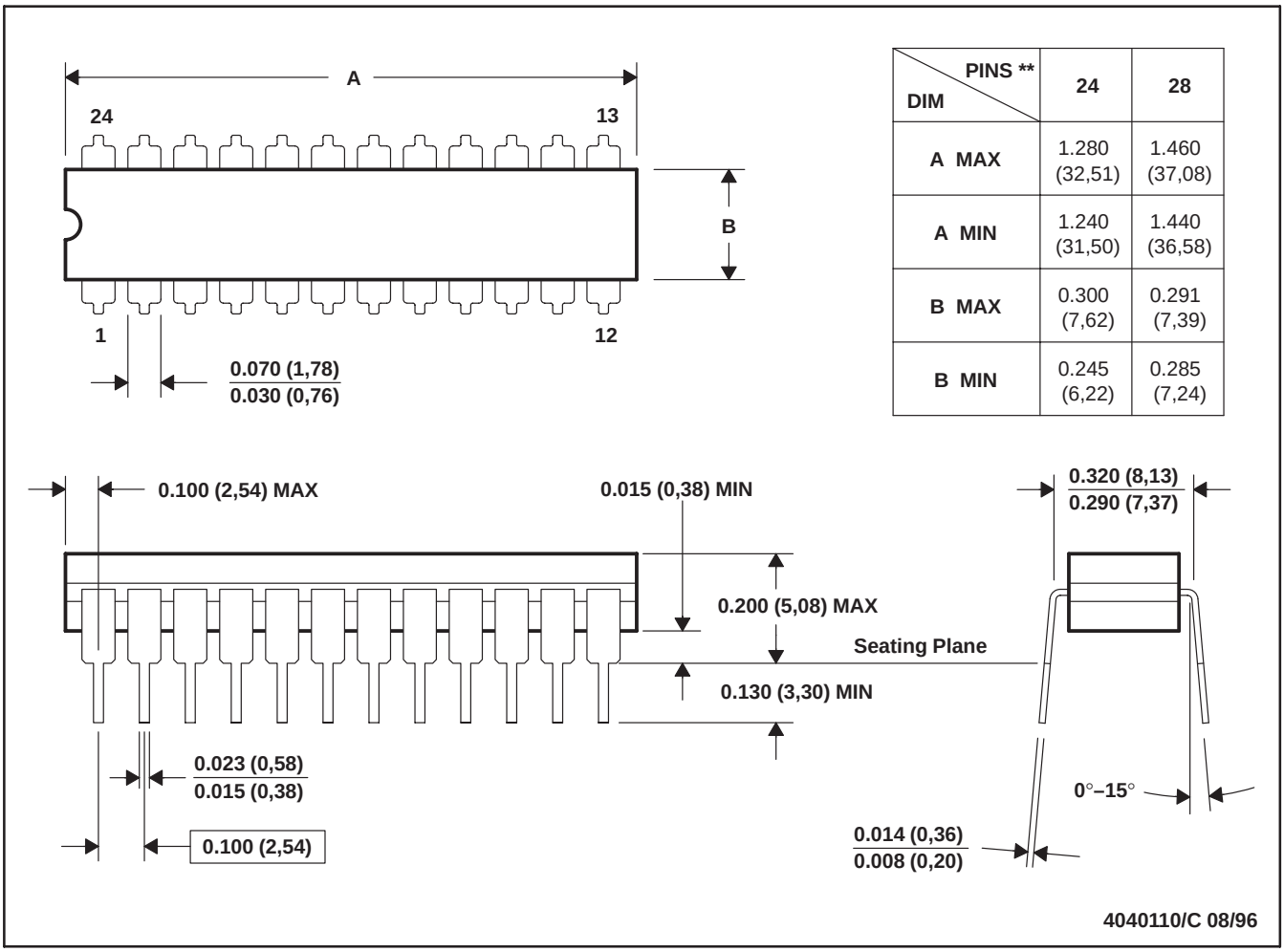
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MCER004A – JANUARY 1995 – REVISED JANUARY 1997

JT (R-GDIP-T)** **CERAMIC DUAL-IN-LINE**

24 LEADS SHOWN



DIM \ PINS **	24	28
A MAX	1.280 (32,51)	1.460 (37,08)
A MIN	1.240 (31,50)	1.440 (36,58)
B MAX	0.300 (7,62)	0.291 (7,39)
B MIN	0.245 (6,22)	0.285 (7,24)

NOTES: A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. This package can be hermetically sealed with a ceramic lid using glass frit.
D. Index point is provided on cap for terminal identification.
E. Falls within MIL STD 1835 GDIP3-T24, GDIP4-T28, and JEDEC MO-058 AA, MO-058 AB

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Last updated 10/2025